

Over-Squashing vs. Adversarial Robustness in GNNs

An “Add vs. Remove Edge” Investigation

Supanut Kompayak (st126055) Dechathon Niamsa-Ard (st126235)

Deep Learning Course Project, Asian Institute of Technology

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The question

Two GNN problems that look unrelated, but both are about **structural sensitivity**:

- **Over-squashing** — long-range signal squeezed through bottlenecks and lost. Output *not sensitive enough* to distant nodes.
- **Adversarial fragility** — a few edited edges flip the prediction. Output *too sensitive* to malicious structural change.

Do methods that fix over-squashing also help robustness?

The spine: add vs. remove edge

- Over-squashing rewiring: FoSR **adds** edges; spectral pruning **removes** edges (both raise the spectral gap).
- Adversarial attack: **adds** dissimilar edges. Defense (GCN-Jaccard): **removes** them.

Hypothesis: *adding* to fix over-squashing may *hurt* robustness (more edges to exploit); *removing* for over-squashing may *transfer* to robustness (removes like a defense).

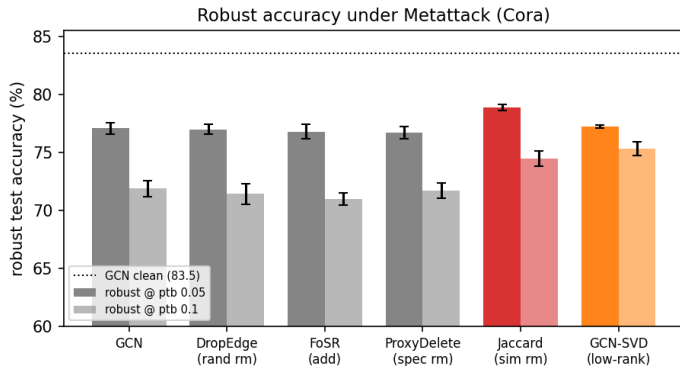
Methods (the design)

Method	Targets	Edge op	Role
GCN (vanilla)	—	none	baseline
FoSR	over-squashing	add	“add” arm
ProxyDelete	over-squashing	remove	“remove” arm
GCN-Jaccard	defense	remove	defense reference
DropEdge	regularization	random	random-removal control

FoSR vs. ProxyDelete = direction; ProxyDelete vs. Jaccard = criterion.

- Cora node classification, DeepRobust `netrack` split (2485 nodes), CPU.
- Attack: Metattack (poisoning), `ptb` {0.05, 0.10}; cached once, reused.
- Pipeline: `clean` $\rightarrow$ Metattack $\rightarrow$ method $\rightarrow$ train GCN $\rightarrow$ eval.
- Metric: clean / robust accuracy, robustness **gap** = clean – robust.

Results: add vs. remove under attack (Cora)



Four arms (GCN, DropEdge, FoSR, ProxyDelete) cluster at the GCN level; **the two purpose-built defenses separate** — Jaccard (similarity) & SVD (low-rank). The add/remove *direction* does not matter — the editing *criterion* does.

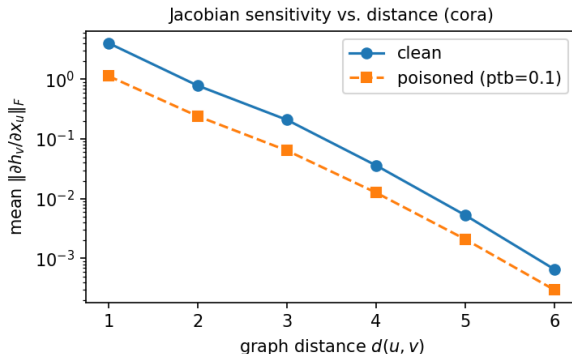
Why does Jaccard win? (mechanism)

Metattack almost purely **adds** edges (503 added, 3 removed at ptb 0.10). How many does each method delete?

Method (ptb .10)	# removed	attack edges rm.	recall
GCN-Jaccard	797	250	0.50
ProxyDelete (spectral)	643	93	0.19
DropEdge (random)	643	59	0.12
FoSR (add)	0	0	0.00

Chance recall ≈ 0.12 . Jaccard deletes **half** the attack edges ($\sim 4\times$ chance) $\Rightarrow$ it undoes the attack.
Spectral/random $\approx$ chance; add removes none.

Over-squashing probe (RQ1)



Sensitivity decays $\sim 5\times/\text{hop} \Rightarrow$ over-squashing is real. Poisoning shifts the level but **not** the decay rate $\Rightarrow$ over-squashing & fragility are decoupled (matches RQ2).

Adaptive attacker: evadable but self-defeating

A Jaccard-aware attacker adds edges between feature-**similar** nodes (the filter won't remove them).

- Evasion works at the filter: Jaccard removes **0%** of these edges.
 - But the attack turns weak: GCN drops only 0.7/2.1 pts (vs 6.5/11.7 for Metattack).
- ⇒ Edges that fool the model (dissimilar) are the ones Jaccard removes; edges that evade Jaccard (similar) barely fool it — **potency & detectability are coupled**.

Heuristic (not gradient-optimal) adaptive attack; optimal adaptive = future work.

Findings

- Adding (FoSR) & spectral removal (ProxyDelete): **no robustness gain** — track the GCN. (FoSR even triples the spectral gap, its own objective — still no gain.)
 - Only **attack-aware defenses** defend: Jaccard (similarity) & SVD (low-rank).
- ⇒ The editing *criterion* matters, not the add/remove *direction*.

An honest negative result for the over-squashing → robustness bridge — real numbers only.

Conclusion

- Over-squashing mitigations (FoSR add, ProxyDelete spectral-remove) **do not** improve adversarial robustness on Cora / Metattack.
- Only attack-aware defenses defend — Jaccard (similarity) & SVD (low-rank); robustness comes from targeting the attacker's edges, not from optimizing the spectral gap.
- Honest negative result: the over-squashing $\rightarrow$ robustness bridge is *not* supported here.
- Robust to budget (spectral removal still fails at 643 edges) & replicated on **Cora + Citeseer**. A defense-aware attacker evades Jaccard only by turning weak.
- Next: gradient-optimal adaptive attack, RQ3 (Jacobian $\rightarrow$ fragility).

Thank you

Questions?